

1. Responsibilities of a Data Engineer:-

- Collect and process data from different sources
- Build and maintain data pipelines (ETL/ELT)
- Store data in DB / data warehouse.
- Ensures data quality / reliability.
- Support data analysts and data scientists

2. Essential skills for a Data Engineering:-

- Programming:- Python, SQL, Java
- ETL Tools:- Data Ingestion Tools
- Big data Tools → Hadoop, Spark
- Database knowledge → SQL & NoSQL
- Cloud platforms → AWS, Azure, GCP.

3) Four principles of effective data architecture:-

1. Scalability → Handle growing data
2. Flexibility → supports different data types
3. Data Quality → Accurate and clean data.
4. Security → Protect sensitive information.

4) Role of data governance:-

- Data governance ensures proper management of data
- Defines rules, policies and standards.
 - Ensures data quality, security and privacy.
 - Controls ~~data~~ access to data.
 - Maintains compliance with regulations.

5. Need for Data Engineering:-

- Organisations generate huge data daily.
- Helps in processing and organising data.
- Supports analytics and AI systems
- Improves decision-making.

6) How data architecture affects ^{data} ~~data~~ quality:-

- proper design ensure clean and consistent data.
- Reduces duplication and errors.
- Standardizes format.
- Improves data validation.

7) Core concepts of data storage systems:-

- Database → stores structured data.
- Data warehouses → Analytics storage
- Data lakes → stores raw data.
- cloud storage → scalable and flexible

8) Draw-backs of Traditional data processing systems:-

- Limited scalability
- Slow processing speed
- Cannot handle unstructured data.
- High maintenance cost
- Lack of real-time processing.

7. Role of Source systems in data pipeline:-
- Source systems generate original data (apps, sensors, websites)
 - Provide input for ETL process.
 - Ensure continuous data flow
- > They are the starting point of data pipelines.

- 10) Data architecture & business strategy alignment:-
- Supports business goals using data
 - Provides insights for decision making
 - Improves efficiency and performance
 - Enables innovation (AI, analytics).

1. How data architecture enables storage & management of structured and unstructured data:-

Ans) Data architecture defines how data is collected, stored and managed.

→ Structured data (tables rows - like databases)

- stored in relational DB.
- Uses schemas (fixed format)

→ Unstructured data (img, video, text, social media)

- stored in NoSQL DB, data lakes or cloud storage
- Flexible format.

→ How architecture helps:

- separate storage layers (data warehouse + data lake)
- Uses proper tools for each layer
- Ensures easy access and management.

2) Integrating multiple heterogeneous data sources:-

Ans):- steps:

1. Data collecting: From APIs, DB, sensors, apps
 2. Data Integration (ETL process):- Extract → Transform → Load.
 3. Data Storage :- Store in data warehouse / lake
 4. Data access layer:- Provides unified access to users
- Tech used:-
 - ① ETL Tools
 - ② Middleware
 - ③ APIs.

3. Data architecture in social media platforms:-

Ans) Key components:-

- User data storage → Profiles, followers (structured)
- Content storage → Img, videos (unstructured).
- Real-Time processing → Like, comments, feeds
- Recommendation systems. → suggests, posts/users

Architecture layers:-

- Frontend → user interface.
- Backend → logic & API's
- DB → store data
- Analytics → user behaviour analysis

4) Data architecture for analytics & insights:-

Ans) Data architecture supports analytics by organising data properly.

Processes:-

1. Data collection from sources
2. Data cleaning & transformation
3. Storage in data warehouse.
4. Analysis using Tools.

Key features:-

- High data quality.
- Fast query processing
- Historical data storage.

5. Importance in data-driven decision making:-

Any) Data architecture ensures:-

- ~~accurated~~ data → Better decisions
- Timely access → Faster decisions
- Consistency access → Reliable Insights

Example:-

- Business uses sales data to improve products
- Hospitals use patient data for better treatment

6) Common data architecture frameworks:-

1.) Data warehouse architecture:

- Central storage for structured data
- Used for reporting & analytics.

2.) Data lake architecture:-

- Storage raw data (structured + unstructured)
- Flexible and scalable

3.) Lambda architecture:-

- Combines batch + real-time processing.

4.) Kappa architecture:-

- Focuses only on real-time data processing.

5.) Data Mesh:-

- Decentralized approach
- Different teams manage their own data.